

- 9 (a) e.g. zero output impedance/resistance
 infinite input impedance/resistance
 infinite (open loop) gain
 infinite bandwidth
 infinite slew rate
1 each, max. 3 B3 [3]
- (b) (i) graph: square wave M1
 correct cross-over points where $V_2 = V_1$ A1
 amplitude 5V A1
 correct polarity (*positive at $t = 0$*) A1 [4]
- (ii) correct symbol for LED M1
 diodes connected correctly between V_{OUT} and earth A1
 correct polarity consistent with graph in (i) A1 [3]
(R points 'down' if (i) correct)
- 10 X-ray images taken from different angles / X-rays directed from different angles B1
 of one section / slice (1)
 all images in the same plane (1)
 images combined to give image of section / slice B1
 images of successive sections / slices combined B1
 image formed using a computer B1
 image formed is 3D image (1)
 that can be rotated / viewed from different angles (1)
(four B-marks plus any two additional marks) B2 [6]

- 11 (a) e.g. noise can be eliminated / filtered / signal can be regenerated